

Saikat Das

Prospective Graduate Researcher in Artificial Intelligence and Computer Vision

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GitHub | LinkedIn | Codeforces | Portfolio

Research Profile

Computer Science and Engineering graduate interested in developing reliable, efficient, and practically deployable artificial intelligence systems. Primary interests include computer vision, medical imaging, vision-language models, explainable AI, efficient deep learning, Edge AI, and large language model adaptation. Research experience includes deployment-oriented neural-network evaluation, medical-image explainability study design, resource-aware AI systems, technical writing, and reproducible experimental planning. Seeking a research-based graduate position where strong engineering skills can support rigorous and socially meaningful AI research.

Research Interests

- Computer Vision and Medical Imaging
- Vision-Language and Multimodal Models
- Explainable and Trustworthy AI
- AI for Clinical Decision Support
- Efficient Deep Learning and Edge AI
- LLM Adaptation and Fine-Tuning
- Model Reliability and Error Detection
- Low-Resource Bangla AI Systems

Education

B.Sc. in Computer Science and Engineering

Graduated 2026

Ahsanullah University of Science and Technology (AUST), Dhaka

Four-year undergraduate program | Total degree requirement: **156 credits**

CGPA: **3.32/4.00** | CGPA over the last 80 credits: **3.65/4.00**

Higher Secondary Certificate (HSC)

2020

Chittagong College

GPA: **5.00/5.00**

Secondary School Certificate (SSC)

2018

Saint Placid's High School

GPA: **5.00/5.00**

Relevant Coursework and Foundations

Advanced Machine Learning, Digital Image Processing, Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Computer Architecture, Software Engineering, System Design, and Mathematics for Computing.

Research Manuscripts

Under Review

1. **Saikat Das**, et al. "A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging."
PLOS ONE, under review; submitted April 2026.

Manuscript in Preparation

2. **Saikat Das** and Tanvir Hasanat. "Do Explanations Agree When the Model Is Right? Cross-Method Saliency Agreement as a Label-Free Error Signal for Chest X-Ray Pneumonia Detection."
Manuscript in preparation.

Research Experience

Edge-AI Waste Classification and Secure Smart-Bin System

GitHub Repository | Demonstration Video

- Developed a deployment-oriented computer-vision system for real-time waste classification on a Raspberry Pi 4B.

- Evaluated six CNN architectures across accuracy, latency, throughput, model size, CPU and memory use, and thermal behavior.
- Integrated TensorFlow Lite inference, OpenCV, biometric authentication, hardware actuation, secure logging, and Firebase synchronization.
- Investigated model selection under resource constraints by jointly considering predictive performance and deployment requirements.

Cross-Method Saliency Agreement for Chest X-Ray Classification

Ongoing

- Investigating whether disagreement among saliency methods can identify classification errors without requiring additional localization labels.
- Focusing on chest x-ray pneumonia classification, explainability reliability, model-error detection, external validation, and uncertainty-aware evaluation.
- Developing the study methodology and evaluation protocol; uncompleted experiments are not presented as results.

Ongoing Research

Large Language Model Fine-Tuning

- Currently studying practical methods for adapting pretrained large language models to task-specific and domain-specific applications.
- Exploring dataset preparation, supervised fine-tuning, parameter-efficient fine-tuning concepts, prompt design, model evaluation, and responsible model use.
- Building the technical foundation required for future research involving language models, multimodal learning, and vision-language systems.

Planned Research

Bangla Sign Language to Bengali Language Conversion and Dataset Development

- Planning a computer-vision system for converting Bangla Sign Language gestures into written Bengali language.
- Intending to develop and document a structured dataset for Bangla Sign Language recognition, subject to data availability, participant consent, and an appropriate collection protocol.
- Proposed research directions include gesture recognition, temporal modelling, sign-to-text conversion, low-resource learning, and accessibility-focused evaluation.
- This project is currently at the planning stage; dataset collection and model experiments have not yet begun.

Technical Skills

Programming	Python, C++, JavaScript, SQL, MATLAB, Java (basic), and PHP (basic)
Machine Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, model evaluation, and data preprocessing
Computer Vision	OpenCV, CNNs, image classification, saliency methods, Vision Transformers, and Vision-Language Models
Language Models	Hugging Face Transformers, LLM concepts, supervised fine-tuning concepts, prompt engineering, and model evaluation
Efficient and Edge AI	TensorFlow Lite, model quantization, model compression concepts, Raspberry Pi, and latency and resource evaluation
Research Tools	Jupyter Notebook, Google Colab, Git, GitHub, Linux, MATLAB, and LaTeX
Backend and Data	Django, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, MySQL, Firebase, and data pipelines

Research and Academic Skills

- Experimental planning, model comparison, quantitative evaluation, and reproducible workflow design.
- Academic writing, literature review, technical documentation, and research presentation.
- Strong foundation in algorithms, software development, database systems, computer networks, and system design.
- Ability to connect machine-learning research with deployable software and resource-constrained hardware.

Academic Service and Peer Review

- Ad Hoc Peer Reviewer, PLOS ONE2026
- Reviewer, Undergraduate Research Symposium, AUST CSE Fest2024

Selected Certifications

- **Vision-Language Models Bootcamp**, OpenCV University, February 2026
- **PyTorch Bootcamp** and **TensorFlow-Keras Bootcamp**, OpenCV University, 2025–2026
- **Machine Learning Onramp** and **Deep Learning Onramp**, MathWorks, July 2026
- **Introduction to Large Language Models**, **Introduction to Generative AI**, **Prompt Design in Vertex AI**, and **Responsible AI**, Google Cloud Skills Boost, 2025–2026
- **Introduction to Data Engineering**, IBM/Coursera, February 2026

Programming and Academic Activities

- Solved **200+** **programming problems** on Codeforces, LeetCode, and other competitive-programming platforms, with emphasis on data structures, algorithms, and analytical problem-solving.
- Contributed reviewer activity to the AUST CSE Fest 2024 undergraduate research symposium.
- Maintains research and software projects through GitHub, with experience in implementation, debugging, deployment, and technical documentation.

Languages

Bengali: Native

English: Intermediate (B1)

Academic References

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